

Ass3

Question 1

1. Analysis of primal problem

The optimization problem is:

$$\begin{array}{ll}\text{minimize} & x^2 + 1 \\ \text{subject to} & (x - 2)(x - 4) \leq 0\end{array}$$

Feasible set:

The constraint $(x - 2)(x - 4) \leq 0$ defines the interval where the product is non-positive. This occurs when x is between the roots 2 and 4:

$$\mathcal{X} = [2, 4]$$

Optimal solution:

The objective $x^2 + 1$ is minimized at the smallest x in the feasible set, which is $x = 2$.

Optimal value:

$$p^* = 2^2 + 1 = 5$$

2. Lagrangian and dual function

The Lagrangian is:

$$L(x, \lambda) = x^2 + 1 + \lambda(x - 2)(x - 4)$$

Plotting:

- Objective $x^2 + 1$ is a parabola with minimum at $x = 0$.
- Feasible set is $[2, 4]$, optimal point at $x = 2$ with value 5.
- For $\lambda \geq 0$, $L(x, \lambda)$ is a quadratic in x . For example:
 - $\lambda = 1$: $L(x, 1) = x^2 + 1 + (x - 2)(x - 4) = 2x^2 - 6x + 9$, minimized at $x = 1.5$ with value 4.5.
 - $\lambda = 2$: $L(x, 2) = 3x^2 - 12x + 17$, minimized at $x = 2$ with value 5.

Lower bound property:

$$p^* = 5 \geq \inf_x L(x, \lambda) \quad \text{for } \lambda \geq 0$$

$$\text{For } \lambda = 1, \inf_x L(x, 1) = 4.5 \leq 5. \quad \text{For } \lambda = 2, \inf_x L(x, 2) = 5 \leq 5.$$

Dual function:

$$g(\lambda) = \inf_x L(x, \lambda)$$

To find $g(\lambda)$, set $\frac{dL}{dx} = 0$:

$$\frac{d}{dx} [x^2 + 1 + \lambda(x^2 - 6x + 8)] = 2x + \lambda(2x - 6) = 0$$

$$x(2 + 2\lambda) - 6\lambda = 0 \implies x = \frac{6\lambda}{2 + 2\lambda} = \frac{3\lambda}{1 + \lambda}$$

Substitute back into $L(x, \lambda)$:

$$g(\lambda) = \left(\frac{3\lambda}{1 + \lambda} \right)^2 + 1 + \lambda \left(\left(\frac{3\lambda}{1 + \lambda} \right)^2 - 6 \left(\frac{3\lambda}{1 + \lambda} \right) + 8 \right)$$

Simplify:

$$\begin{aligned} g(\lambda) &= \frac{9\lambda^2}{(1 + \lambda)^2} + 1 + \lambda \left(\frac{9\lambda^2 - 18\lambda(1 + \lambda) + 8(1 + \lambda)^2}{(1 + \lambda)^2} \right) \\ &= \frac{9\lambda^2 + (1 + \lambda)^2 + \lambda(9\lambda^2 - 18\lambda - 18\lambda^2 + 8 + 16\lambda + 8\lambda^2)}{(1 + \lambda)^2} \\ &= \frac{9\lambda^2 + 1 + 2\lambda + \lambda^2 + \lambda(-\lambda^2 - 2\lambda + 8)}{(1 + \lambda)^2} \\ &= \frac{-\lambda^3 + 8\lambda^2 + 8\lambda + 1}{(1 + \lambda)^2} \end{aligned}$$

Sketch:

- $g(\lambda)$ is concave in $\lambda \geq 0$.
- At $\lambda = 0$: $g(0) = 1$.
- At $\lambda \rightarrow \infty$: $g(\lambda) \approx -\lambda + 10$.

3. Lagrange dual problem

The dual problem is:

$$\text{maximize } g(\lambda) \text{ subject to } \lambda \geq 0$$

Optimal dual solution:

To find λ^* , set $\frac{dg}{d\lambda} = 0$:

$$\frac{dg}{d\lambda} = \frac{(-3\lambda^2 + 16\lambda + 8)(1 + \lambda)^2 - (-\lambda^3 + 8\lambda^2 + 8\lambda + 1)(2)(1 + \lambda)}{(1 + \lambda)^4} = 0$$

Numerically solving $\frac{dg}{d\lambda} = 0$ for $\lambda \geq 0$ gives $\lambda^* \approx 2$.

Strong duality:

Since the primal is convex and Slater's condition holds (e.g., $x = 3$ is strictly feasible), strong duality holds:

$$p^* = d^* = 5$$

4. Sensitivity analysis

The perturbed problem:

$$p^*(u) = \inf\{x^2 + 1 \mid (x - 2)(x - 4) \leq u\}$$

For $u = 0$, $p^*(0) = 5$. For $u > 0$, the feasible set expands, and $p^*(u)$ decreases. For $u < 0$, the feasible set shrinks, and $p^*(u)$ increases.

Derivative:

$$\frac{dp^*(0)}{du} = -\lambda^* \approx -2$$

Question 2

The primal problem is:

$$\begin{aligned} &\text{minimize} && c^T x \\ &\text{subject to} && f(x) \leq 0 \end{aligned}$$

The Lagrangian is:

$$L(x, \lambda) = c^T x + \lambda f(x)$$

The dual function is:

$$g(\lambda) = \inf_x \{c^T x + \lambda f(x)\} = -\sup_x \{-c^T x - \lambda f(x)\} = -f^*(-c, \lambda)$$

where $f^*(z, \lambda)$ is the conjugate of f with respect to x .

The dual problem is:

$$\text{maximize } -f^*(-c, \lambda) \text{ subject to } \lambda \geq 0$$

Convexity:

The dual problem is convex because f^* is convex (as the conjugate of any function is convex), and maximizing $-f^*$ is equivalent to minimizing f^* , which is convex.

Question 3

The primal LP is:

$$\begin{aligned} &\text{minimize} && c^T x \\ &\text{subject to} && Gx \preceq h \\ &&& Ax = b \end{aligned}$$

The Lagrangian is:

$$L(x, \lambda, \nu) = c^T x + \lambda^T (Gx - h) + \nu^T (Ax - b)$$

The dual function is:

$$g(\lambda, \nu) = \inf_x \{c^T x + \lambda^T Gx + \nu^T Ax - \lambda^T h - \nu^T b\}$$

For the infimum to be finite, the coefficient of x must be zero:

$$c + G^T \lambda + A^T \nu = 0$$

Thus:

$$g(\lambda, \nu) = -\lambda^T h - \nu^T b \quad \text{if} \quad c + G^T \lambda + A^T \nu = 0$$

The dual problem is:

$$\begin{aligned} & \text{maximize} && -h^T \lambda - b^T \nu \\ & \text{subject to} && G^T \lambda + A^T \nu + c = 0 \\ & && \lambda \succeq 0 \end{aligned}$$

Implicit equality constraints:

The equality $G^T \lambda + A^T \nu + c = 0$ is explicit. The implicit constraints are $\lambda \geq 0$.

Question 4

The primal problem is:

$$\text{minimize} \quad \max_{i=1, \dots, m} (a_i^T x + b_i)$$

1. Dual form

Introduce $y_i = a_i^T x + b_i$:

$$\begin{aligned} & \text{minimize} \quad \max_i y_i \\ & \text{subject to} \quad a_i^T x + b_i = y_i, \quad i = 1, \dots, m \end{aligned}$$

The Lagrangian is:

$$L(x, y, \nu) = \max_i y_i + \sum_{i=1}^m \nu_i (a_i^T x + b_i - y_i)$$

The dual function is:

$$g(\nu) = \inf_{x, y} \left\{ \max_i y_i + \sum_{i=1}^m \nu_i (a_i^T x + b_i - y_i) \right\}$$

For the infimum to be finite, $\sum \nu_i a_i = 0$ and $\sum \nu_i = 1, \nu_i \geq 0$:

$$g(\nu) = \sum_{i=1}^m \nu_i b_i \quad \text{if} \quad \sum \nu_i a_i = 0, \sum \nu_i = 1, \nu_i \geq 0$$

The dual problem is:

$$\begin{aligned} & \text{maximize} \quad \sum_{i=1}^m \nu_i b_i \\ & \text{subject to} \quad \sum_{i=1}^m \nu_i a_i = 0 \\ & && \sum_{i=1}^m \nu_i = 1 \\ & && \nu_i \geq 0, \quad i = 1, \dots, m \end{aligned}$$

2. LP formulation

Introduce t such that $a_i^T x + b_i \leq t$ for all i :

$$\begin{aligned} & \text{minimize} && t \\ & \text{subject to} && a_i^T x + b_i \leq t, \quad i = 1, \dots, m \end{aligned}$$

The dual of this LP is:

$$\begin{aligned} & \text{maximize} && \sum_{i=1}^m \nu_i b_i \\ & \text{subject to} && \sum_{i=1}^m \nu_i a_i = 0 \\ & && \sum_{i=1}^m \nu_i = 1 \\ & && \nu_i \geq 0, \quad i = 1, \dots, m \end{aligned}$$

This matches the dual derived in part 1.

Question 5

The primal problem is:

$$\text{minimize} \quad - \sum_{i=1}^m \log(b_i - a_i^T x) \quad \text{with domain } \{x \mid a_i^T x < b_i\}$$

Introduce $y_i = b_i - a_i^T x$:

$$\begin{aligned} & \text{minimize} && - \sum_{i=1}^m \log y_i \\ & \text{subject to} && y_i = b_i - a_i^T x, \quad i = 1, \dots, m \end{aligned}$$

The Lagrangian is:

$$L(x, y, \nu) = - \sum_{i=1}^m \log y_i + \sum_{i=1}^m \nu_i (b_i - a_i^T x - y_i)$$

The dual function is:

$$g(\nu) = \inf_{x, y} \left\{ - \sum \log y_i + \sum \nu_i (b_i - a_i^T x - y_i) \right\}$$

For the infimum over x to be finite, $\sum \nu_i a_i = 0$. For y_i :

$$\frac{\partial}{\partial y_i} (-\log y_i - \nu_i y_i) = -\frac{1}{y_i} - \nu_i = 0 \implies y_i = -\frac{1}{\nu_i}$$

Thus:

$$g(\nu) = - \sum \log \left(-\frac{1}{\nu_i} \right) + \sum \nu_i \left(b_i - \frac{1}{\nu_i} \right) = \sum (\log \nu_i + 1 + \nu_i b_i - 1) = \sum (\log \nu_i + \nu_i b_i)$$

subject to $\sum \nu_i a_i = 0$, $\nu_i \leq 0$ (since $y_i > 0$ and $y_i = -1/\nu_i$).

The dual problem is:

$$\begin{aligned} & \text{maximize} && \sum_{i=1}^m (\log \nu_i + \nu_i b_i) \\ & \text{subject to} && \sum_{i=1}^m \nu_i a_i = 0 \\ & && \nu_i \leq 0, \quad i = 1, \dots, m \end{aligned}$$

Question 6

The primal problem is:

$$\begin{aligned} & \text{minimize} && e^{-x} \\ & \text{subject to} && \frac{x^2}{y} \leq 0 \\ & && y > 0 \end{aligned}$$

1. Convexity and optimal solution

- The objective e^{-x} is convex.
- The constraint $\frac{x^2}{y} \leq 0$ is convex since x^2/y is convex for $y > 0$.
- The constraint implies $x^2 \leq 0$, so $x = 0$.
- The optimal solution is $x = 0, y > 0$ (any $y > 0$ is optimal).
- The optimal value is $p^* = e^0 = 1$.

2. Lagrange dual problem

The Lagrangian is:

$$L(x, y, \lambda) = e^{-x} + \lambda \frac{x^2}{y}$$

The dual function is:

$$g(\lambda) = \inf_{x, y > 0} \left\{ e^{-x} + \lambda \frac{x^2}{y} \right\}$$

For $\lambda = 0, g(0) = 0$. For $\lambda > 0$, the infimum over y is $y \rightarrow 0^+$, giving $g(\lambda) = -\infty$. Thus:

$$g(\lambda) = \begin{cases} 0 & \text{if } \lambda = 0 \\ -\infty & \text{if } \lambda > 0 \end{cases}$$

The dual optimal is $\lambda^* = 0, d^* = 0$.

Optimal duality gap:

$$p^* - d^* = 1 - 0 = 1$$

3. Slater's condition

Slater's condition requires a strictly feasible point where $\frac{x^2}{y} < 0$. However, $\frac{x^2}{y} \leq 0$ implies $x = 0$, so no strictly feasible point exists. Thus, Slater's condition does not hold.

4. Perturbed problem

The perturbed problem is:

$$p(u) = \inf \{ e^{-x} \mid \frac{x^2}{y} \leq u, y > 0 \}$$

- For $u < 0$, no feasible x exists, so $p(u) = +\infty$.
- For $u = 0, x = 0, p(0) = 1$.

- For $u > 0$, x can be non-zero, and the infimum is $p(u) = 0$ (achieved as $x \rightarrow +\infty$).

Thus:

$$p(u) = \begin{cases} +\infty & \text{if } u < 0 \\ 1 & \text{if } u = 0 \\ 0 & \text{if } u > 0 \end{cases}$$